Product

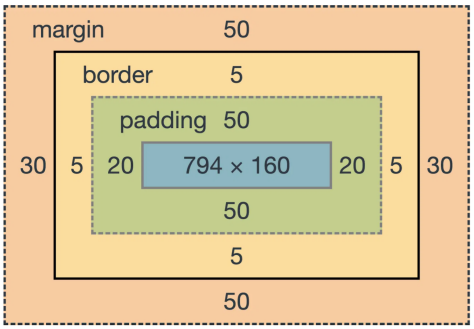
```
function filter<T>(elems:T[], predicateFn) {
    let result = [];
    for(var i = 0; i < elems.length; i++) {
        if(predicateFn(elems[i])) {
            result.push(elems[i]);
        }
    }
    return result;
}
```

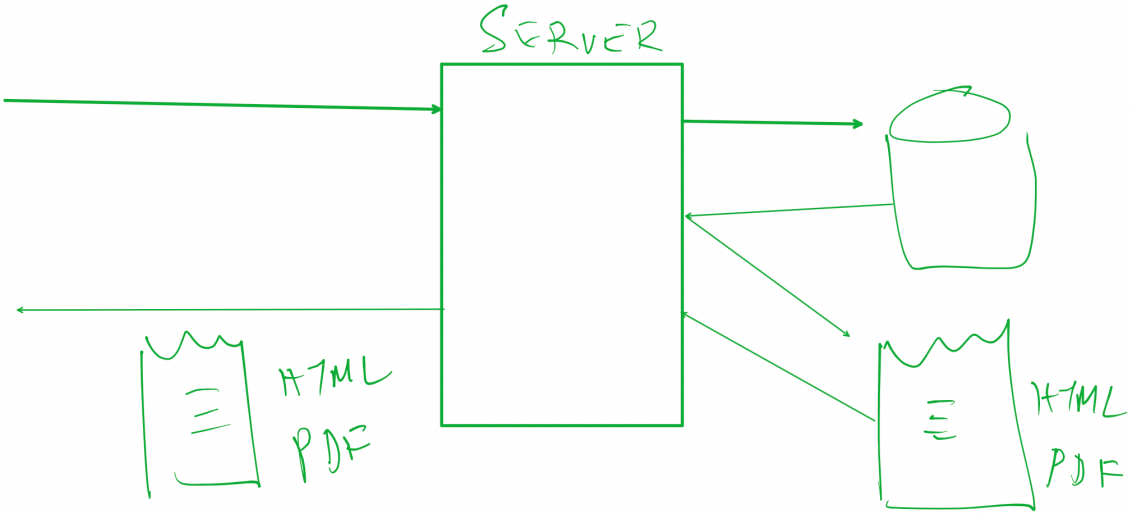
T[ ]

number[ ]

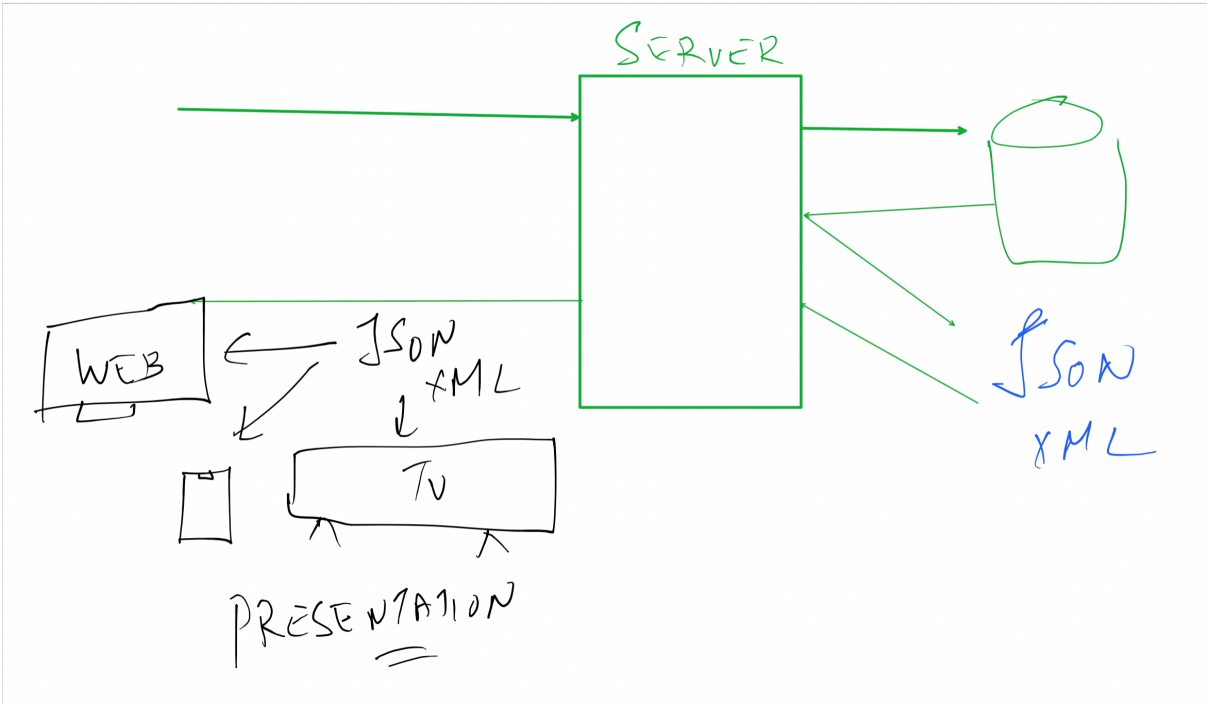
number

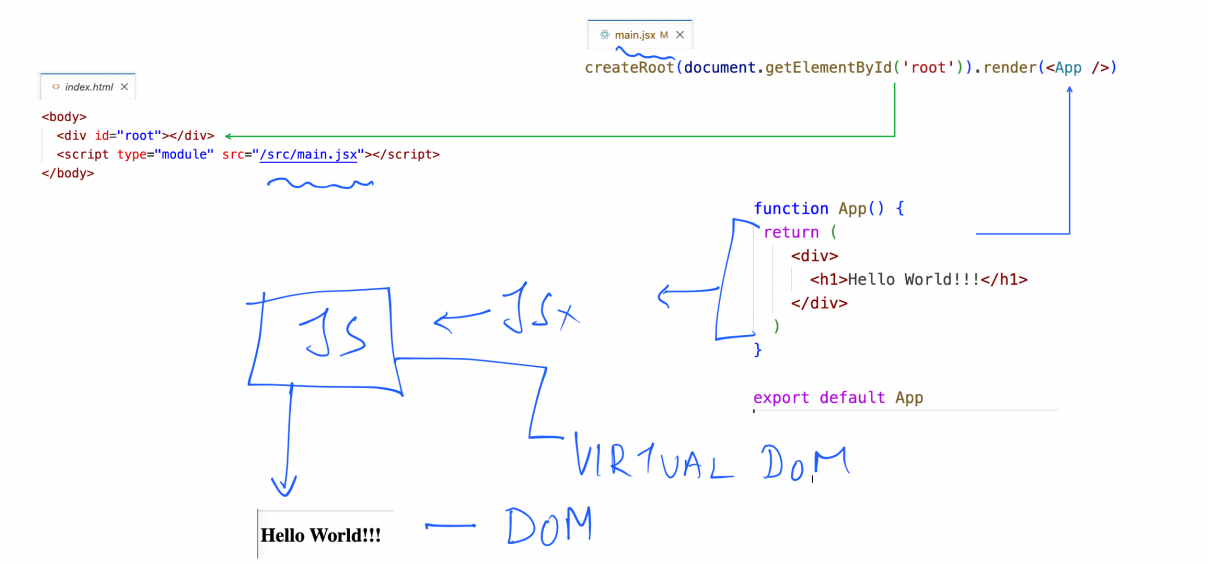
```
filter(products, (p) => p.category === 'Electronics' );

filter(data, (no) => no % 2 === 0);
```





Client Side Rendering:


SERVER
WEB
JSON
XML
TV
PRESENTATION
JSON
XML


main.jsx M
createRoot(document.getElementById('root')).render(<App />)
index.html
<body>

<script type="module" src="/src/main.jsx"></script>
</body>
function App() {
return (

<h1>Hello World!!!</h1>

)
}
export default App
JS
JSX
VIRTUAL DOM
Hello World!!!
DOM

main.jsx M

```jsx
createRoot(document.getElementById('root')).render(<App />)
```

index.html

```html
<body>
  <div id="root"></div>
  <script type="module" src="/src/main.jsx"></script>
</body>
```

```jsx
function App() {
  // bad code
  let product = {
    "name" : "iPhone 17",
    "price": 120933.33
  }
  return (
    <div>
      <h1>{product.name}</h1>
      <p>Price: {product.price}</p>
    </div>
  )
}

export default App
```

INTERPOLATION

JS ← JSX

VIRTUAL DOM

iPhone 17 — DOM

Price: 120933.33

```jsx
function App() {
  return (
    <div>
      <Users />
    </div>
  )
}
```

```jsx
export default function Users() {
    // bad code
    let users = […
    ];
    return (
        <div>
            {
                users.map( user => <User user={user} a={10} b={100}/>)
            }
        </div>
    )
}
```

```jsx
let users = [
  {
    "id": 1,
    "name": "Leanne Graham",
    "username": "Bret",
    "email": "Sincere@april.biz",
    "address": {…
    },
    "phone": "1-770-736-8031 x56442",
    "website": "hildegard.org",
    "company": {…
    }
  },
  {
    "id": 2,
    "name": "Ervin Howell",
    "username": "Antonette",
```

```jsx
export default function User(props) {
    let user = props.user;
    let {name, username, email} = user;
    return (
        <div>
            <h1>{name}</h1>
            <h3>{username}, {email}</h3>
            <hr />
        </div>
    )
}
```

Leanne Graham 1

Bret, Sincere@april.biz

Ervin Howell 2

Antonette, Shanna@melissa.tv